ENGINEERING SCIENCE

Revised Nov 2021

Pupil worksheet

Energy calculations

Topic 1: Calculating energy: Electrical, Kinetic and Potential

Topic 2: Calculating power

Topic 3: Energy loss and transfer

Topic 4: Energy efficiency and audits



Name: _____

Class: _____

Teacher: _____

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Learning Intentions

- To understand what energy is and how it can be used.
- To know how to calculate the energy in different scenarios.
- To learn about energy conservation.
- To learn about the transfer of energy.
- To learn about the efficiency of a system and how to calculate it.
- To know what an energy audit is and why it is used.

Success Criteria

- I can solve energy problems using energy formulae for potential, kinetic, heat, and electrical energy
- Having gained knowledge of how formulae may be used in the context of energy transfer, I can apply them to solve problems, for example in engineering.
- I can use power and work done formulae
- I can identify the energy transfers that happen in a system.
- I can carry out an energy audit.
- I can calculate the efficiency of a system.

Summary of topic

Energy is a topic that permeates all the others in the subject. It affects everything we do every moment of every day, and therefore it is important that the students learn to understand, identify and calculate energy in some of its many forms.

When students have completed this unit of work, they should be able to:

- Understand and identify the many ways that energy affects our lives.
- Describe the environmental implications of using or producing certain forms of energy.
- Identify different forms of renewable energy sources.
- Perform various calculations to determine how much energy is present in a system.
- Produce systems diagrams to show energy transformations.
- Determine how efficient an energy system is.

Success Criteria

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red:	I need to work on this topic further to understand it.
Amber:	I need a bit of help to fully understand this topic.
Green:	I am confident in my understanding of this topic.

- ☐ I can solve energy problems using energy formulae for potential, kinetic, heat, and electrical energy.
- ☐ Having gained knowledge of how formulae may be used in the context of energy transfer, I can apply them to solve problems, for example in engineering.
- ☐ I can use power and work done formulae.
- ☐ I can identify the energy transfers that happen in a system.
- ☐ I can carry out an energy audit.
- ☐ I can calculate the efficiency of a system.

Relationships from Data Booklet

These are the formulae you will be provided with throughout the course and assessments. While you do not need to memorise them, you should be able to recall them and state what each means.

Energy and power

Potential energy	$E_p = mgh$	$g = 9{\cdot}8\ ms^{-2}$ (to 2 significant figures)
Kinetic energy	$E_k = \frac{1}{2}mv^2$	
Heat energy	$E_h = cm\Delta T$	$c_{water} = 4180 Jkg^{-1}K^{-1}$
Electrical energy	$E_e = VIt$	
Work done	$E_w = Fd$	
Power	$P = \frac{E}{t}$	
Electrical power	$P = VI = \frac{V^2}{R} = I^2R$	
Mechanical power	$P = Fv$	$P = 2\pi nT$ (n= no of revs per second)
Efficiency	$\eta = \frac{Energy_{out}}{Energy_{in}} = \frac{Power_{out}}{Power_{in}}$	

Prefixes and Scientific notation

Throughout this course, you should be able to recognise the different prefixes to units and be able to covert to the standard form of the unit for use in formulae. You can do this either by using scientific notation or by multiplying or dividing by the correct factor.

prefix	*symbol*	*notation*	*operation*
nano	n	x 10^{-9}	÷ 1 000 000 000
micro		x 10^{-6}	÷ 1 000 000
milli	m	x 10^{-3}	÷ 1000
kilo	k	x 10^{3}	x 1000
mega	M	x 10^{6}	x 1000 000
giga	G	x 10^{9}	x 1000 000 000

Rounding and significant figures

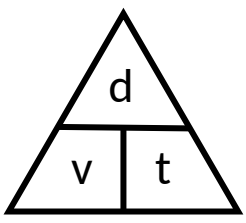
When answering calculations on this course, make sure you are using significant figures within your final answer. In a number, all figures are significant, except zeros at the front (to show decimal points) or at the end. For example, 1234 has four significant figures. 0.012 has only two significant figures, as does 1200. You should also take care to round your numerical answers to an appropriate number of significant figures.

$3.6 \times 10^{3} = 3600.00$ $3.6 \times 10^{-3} = 0.0036$



An electrical signal travels down a copper wire at 3.0×10^{8} ms^{-1}. How long would it take to cover a distance of 830km?

$v = 3.0 \times 10^{8}$ $ms^{-1} = 300\ 000\ 000$



d = 830 km = 830 000m

t = d ÷ v

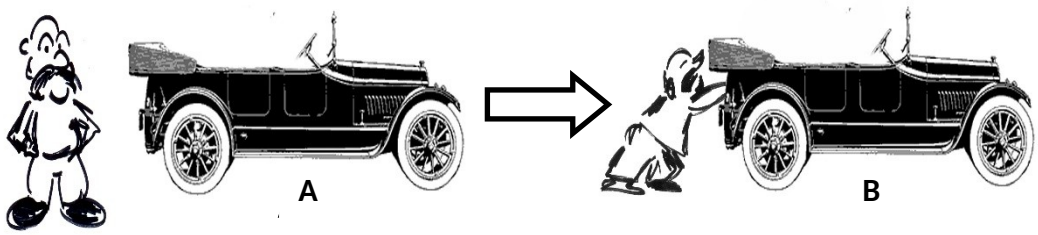
$$t = \frac{830\ 000}{300\ 000\ 000}$$

t = 0.002766s

t = 0.0028s

Work done and Energy

When a force is used to move an object, 'work' is said to be done. An example of this is car which has run out of energy and must be pushed from position A to position B.

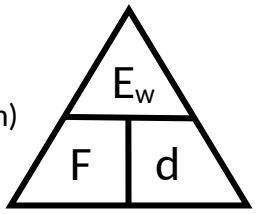


The amount of work you will do depends on how difficult the car is to push (the size of the force) and how far it has to be pushed (the distance). The amount of work that has to be done can be calculated using the formula:

$$E_w = Fd$$

Energy Work done = Force applied x distance moved

- Force (F) is measured in Newtons (N)
- Distance (d) is measured in metres (m)
- Energy Work (E_w) done is measured in Joules (J) or Nm Newton metre (Nm)



Example

A lift raises a mass of 1000kg to the next floor in the Union Square shopping centre, which is 20m up. Calculate the minimum amount of work that must be done by the winch.

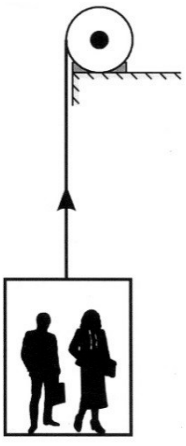
Remember – Gravitational field strength is always 9.8 N/kg. The gravitational field strength is provided in the data booklet as 9.8 ms^{-2} (don't worry about the units as it is the same).

Convert the weight of lift into a force = mg (mass x gravitational field strength)

What are my knowns?
m = 1000kg
d = 20m
g = 9.8 N/kg
F = ?

F = mg
= 1000 x 9.8
= 9800 N

E_w = Force x distance
= 9800 x 20
= **196,000 J or 196KJ**



1. Calculate the amount of work done when lifting the following masses to a height of 2m:

a) 0.5kg b) 2. 1kg c) 3. 2kg d) 3.5kg

2. Calculate the amount of work done when a force of 150N is used to pull a 50kg bag of sand 20 metres.

3. During the loading process, a forklift truck lifts a pallet of bricks with a mass of 740kg to a height of 2m.

a) Calculate the minimum amount of work the truck must do during the lift.

b) Can you also suggest why the actual work done will be greater than this?

4. A shopping trolley is pushed for a distance of 25m. The average force applied is 400N. Calculate the work done.

5. Gardener Jones mows his lawn using a lawnmower with an average force of 230N over a distance of 42m. Calculate the work done.

6. A car is brought to rest over a distance of 24m. The frictional force is 980N. Calculate the work done in bringing the car to rest.

7. A mother pushes her baby in a pram for a distance of 768m. The total word done is 210kJ. Calculate the average force applied.

8. A car tows a caravan for a distance of 12km. The total work done is 30MJ. Calculate the average force applied.

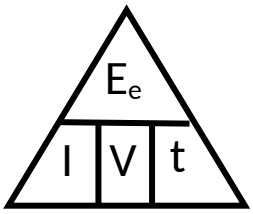
9. A tug boat is towing a small oil rig. The boat produces 205MJ of work by exerting an average force of 15kN on the oil rig. Calculate the distance the oil rig is towed.

10. A person of mass 50kg runs up a flight of stairs. The vertical height of the stairs is 8m. Calculate

a) The weight of the person

b) Work done

Calculating Electrical energy, Ee



The formula for calculation electrical energy is:

$$E_e = IVt$$

Electrical Energy = Current x Voltage x time

- Current (I) is measured in Amps (A)
- Voltage (V) is measured in Volts
- Time (t) is measured in seconds (s)

Example

An electric hob in home economics has an operating voltage of 230 volts with a current of 5 amps. Calculate how much electrical energy has been used if the hob takes 5 minutes to heat up a pot of soup.

Remember to always convert your time into seconds.
Therefore 5 minutes x 60 = 300 seconds.

What are my knowns?
Ee = IVt I = 5A V = 230V t = 300 seconds

= 5 x 230 x 300

= 345,000 Joules

= 345 kJ

11. Find out how much electrical energy is transferred from a 230 volt power supply to a 12 volt battery which draws 5 amps for 2 minutes.

12. A hot air hand drier activates for 45 seconds once the switch is pressed. The drier operates from a 230V supply and draws a current of 10A. Calculate the amount of electrical energy used when the drier is operating.

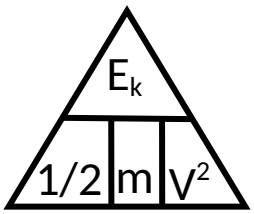
13. A turbine is used to drive an 110V electrical generator. If the output current is 3A, how much electrical energy does the generator produce in 1 hour?

Calculating Kinetic Energy, E_k

The kinetic energy of a moving object is dependent on 2 factors; the mass (m) of the object & its velocity (v)

$$E_k = ½ mv^2$$

Kinetic energy = ½ x mass x velocity^2



- Mass of the object is measured in kg
- Velocity is measured in m/s or ms^{-1} (either one is accepted as they are the same unit of measurement)

Example

The total mass of a go-kart and the driver is 90kg. The go-kart is travelling at 40 m/s, how much kinetic energy does it possess?

What are my knowns?
m = 90kg v = 40 m/s

$E_k = ½ mv^2$

$= ½ \times 90 \times 40^2$

$= 0.5 \times 90 \times 1600$

$= 72{,}000$ joules

$= \underline{72\ kJ}$



14. The first driver of a go-kart had a mass of 50kg and travelled a top speed of 40m/s. They were replaced by a driver with a mass of 70kg. Calculate the kinetic energy both drivers in the go-kart possess when travelling at the same speed. Comment on the difference you find and why it occurs.

15. A crane raises a load of 200kg to a height of 30m in one minute. Determine the kinetic energy of the load when it's moving.

16. A racing car drives around a circuit at a constant speed of 50m/s. If the car has a kinetic energy of 500kJ, what is the mass of the car?

17. A plane has a mass of 9000kg. It flies at a speed of 100m/s. Calculate the kinetic energy of the plane.

18. A runner crosses the finishing line at a speed of 7m/s. At this point the runner has a kinetic energy of 1862J. Calculate the mass of the runner.

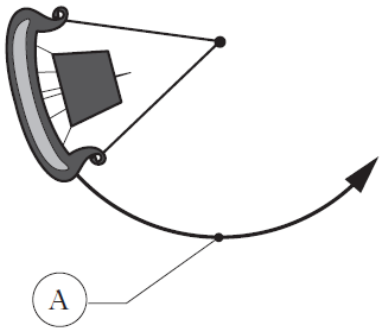
19. A remote controlled car has a speed of 1.2m/s. It has a kinetic energy of 0.216J. Calculate the mass of the car.

20. A lorry of mass 2800kg is travelling at a constant speed. It has a kinetic energy of 132kJ. Calculate the speed of the lorry.

21. A motorbike has a mass of 250kg and is travelling at 18m/s. A car has a mass of 950kg and travels at 12.5m/s. Calculate which has the greater kinetic energy.

22. A car of mass 850kg is travelling at a speed of 20m/s. The brakes are applied and the car reduces to a speed of 12m/s. Calculate the change is the car's kinetic energy.

23. A fairground swing is shown.



The swing and passengers have a combined mass of 2500kg.

(a) Calculate, showing all working and units:

(i) The kinetic energy at its lowest point when the velocity is 15 m/s;

(ii) The maximum height the swing will reach. (Assume no loss of energy). Note: You must use the potential energy equation to solve this problem: $E_p=mgh$. The next theory page explains calculating E_p.

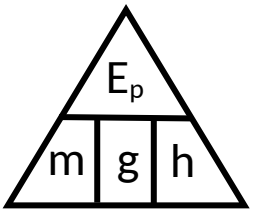
(b) In reality, there will be some loss of energy in the swing.

(i) State one possible cause of energy loss.

(ii) State how this energy loss could be reduced.

Calculating Potential energy Ep

The formula most commonly used to calculate potential energy is:



$$E_P = mgh$$

Potential energy = mass x gravity x height

m is the mass of the object, *g* is the gravitational field strength acting on the object and *h* is the height the object is above the ground or datum.

- Mass of the object and is measured in kilograms (kg)
- Height of the object above the ground or datum is measured in metres (m)
- Remember – the gravitational field strength is always 9.8 N/kg or ms^{-2} on Earth.

Example

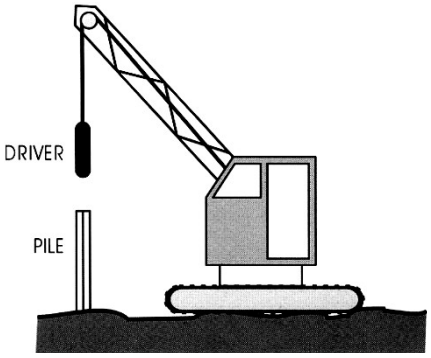
Metal piles are driven into the ground using a pile driver. This consists of a 500 kg driver, which is raised by a winch to a height of 3 m, then released. Calculate the potential energy stored when the driver is at its maximum height.

What are my knowns?
Ep = mgh m = 500kg h = 3m g = 9.8 N/kg

$E_p = 500 \times 9.8 \times 3$

= 14, 700 Joules

= 14.7kJ



24. A 1.5kg toy buggy is set up to roll down ramps of different height to test its potential energy. Calculate how much energy the buggy possesses at the top of each of the following ramp heights:

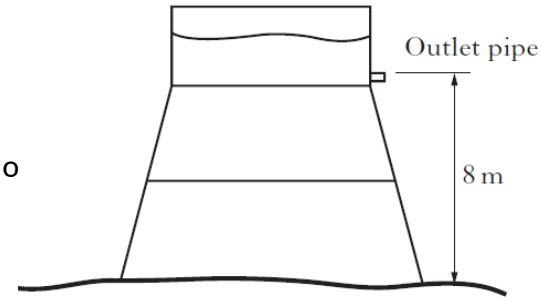
a) 50 cm b) 75 cm c) 1 m d) 1.24 m

25. Using a ramp height of 1.22m, calculate the potential energy of the buggy with different weights.

a) 2.5 kg b) 3.5 kg c) 6kg d) 400g

26. A roller coaster has many high and low points on its track. The height at the top of its largest dip is 50m, and the lowest point is 5m, what is the change in potential energy for a person with a mass 80kg between the highest and lowest points?

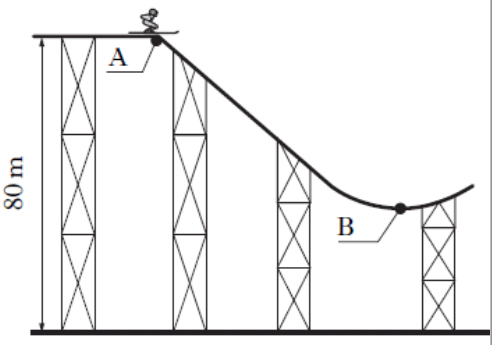
27. A crofter uses a wind power system to pump water to a storage tank. A simplified diagram of the storage tank is shown.



a) Calculate the potential energy of 6L (litres) of water at the o

b) Describe why not all of the water's potential energy is converted.

28. A ski jumper and take-off ramp is shown. The ski jumper has a mass of 65kg.



a) Calculate, showing all working and units, the potential energy of the ski jumper when stationary at point A.

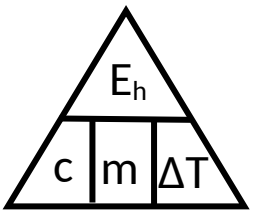
b) The ski jumper descends down the ramp and has a potential energy of 31·5 kJ at point B.

(i) Calculate the kinetic energy of the ski jumper at point B (assume no loss of energy);

(ii) The velocity of the ski jumper at point B.

Calculating Heat Energy

The formula for calculating heat energy is:



$$E_h = cm\Delta T$$

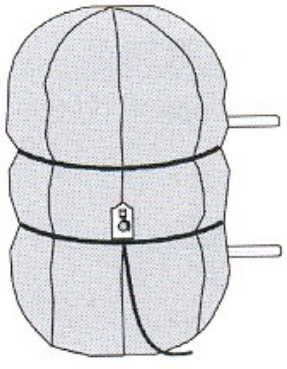
Heat Energy = heat capacity of material x mass x change in temp

- Specific heat capacity (c) is measured in Joules per Kilogram per Kelvin (or Celcius) (J/kgK or J/kg°C)
- Change in temperature (ΔT) ('Delta T') is measured in Celsius (°C) or Kelvin (K)

Important: The specific heat capacity of a substance, is the amount of energy required to raise the temperature of 1kg of the material by 1 degree celsius.

Example

A hot water tank contains 200 litres of water at 18°C. Calculate how much energy is required to raise the temperature of the water to 50°C. 1 litre of water has a mass



of 1kg. The specific heat capacity of water is 4180 J/kgK. The specific heat capacity of a liquid changes depending on the type of liquid. The provided data booklet will specify what value to use.

m = 200L

C_{water} = 4.180 KJ/KgK or 4180 J/KgK (data booklet)

Change in temperature (ΔT) = T2 – T1

= 50 – 18

= 32°C

E = cmΔT

= 4180 x 200 x 32

= 26752000 J

= 26.752 MJ 27MJ (2 s.f)

29. Using a kettle with 1.5 Litre of water and a thermometer, work out the heat energy needed to raise the temperature from 5°C to:

a) 50°C b) 40°C c) 80°C d) 100°C

30. Calculate the heat energy required to heat 1.75kg of water from a temperature of 18°C until it begins to boil.

31. 57kJ of thermal (heat) energy is supplied to 1.7kg of oil having a specific heat capacity of 2.7 kJ/kgK. If the initial temperature of the oil is 3°C, what will be its final temperature?

Electrical Power, P

Whenever there is current in a wire, electrical energy in transformed into another form. This could be heat, light, kinetic energy etc, depending on the electrical appliance being used. Sometimes these transformations are intentional (light) and sometimes they are not (loss energy as heat). Power is a measure of the rate of energy transfer in one second, i.e. the energy transferred in one second. It gives an indication of how quickly the energy is changed from one form to another.

Power = Energy transfer ÷ time

$$P = \frac{E}{t}$$

The energy used in this equation can be in any form, from work done (J), kinetic (J), electrical (J) or indeed any type of energy - providing it is measured in joules.

- Power is measure in watts (W). Either unit is acceptable as P is equivalent to one joule per second (Js^{-1})
- Energy transfer is measured in joules (J)
- Time is measured in seconds (s)

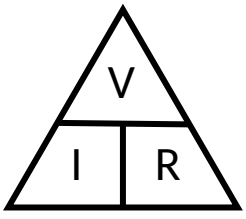
Resistance formula using Ohm's law

$$V = IR$$

V = Potential difference (V)

I = Current (A)

R = Resistance (Ω)



To calculate Energy (E) = P x t, power (P) can be replaced with the value of IV from Ohms law as shown:

Energy (E) = P x t

Therefore Energy (E) = IV x t The Power formulae however is actual written as **P = IV**

Power Relationships (Formulae)

In addition to the definition for power above, there are three other relationships which allow us to calculate the power of an electrical component or appliance. These formulae relate power to potential difference (voltage), current and resistance.

$$P = IV$$

P = power (watts, W)
I = current (amperes, A)
V = potential difference (volts, V)

$$P = \frac{V^2}{R}$$

P = power (watts, W)
V = potential difference (volts, V)
R = resistance (ohms, Ω)

$$P = I^2R$$

P = power (watts, W)
I = current (amperes, A)
R = resistance (ohms, Ω)

Example

If an electric light bulb uses 60 kJ of energy in 10 minutes, what is the power rating of the bulb?



E = 60,000 joules

t = 10 minutes x 60 seconds, therefore t = 600 seconds

$$P = \frac{60\,000}{600}$$

P = 100 Watts

32. A 3.0 kW electric fire is switched on for 2.5 hours. Calculate the energy used in this time.

33. A TV connected to the mains supply (230 V) operates on a current of 1.5 A. Calculate the power rating of the TV.

34. A section of overhead electricity line has a power loss of 14 MW. The current in the cable is 900 A. Calculate the resistance of the cables.

35. A hairdryer used 86.4kJ of energy in a time of 1.5 minutes. Calculate the power rating of the hairdryer.

Conservation of Energy

In everyday usage, energy conservation means using less energy to do the same work. Examples of this include improving the heat insulation of buildings and homes, making cars which use fuel more efficiently…

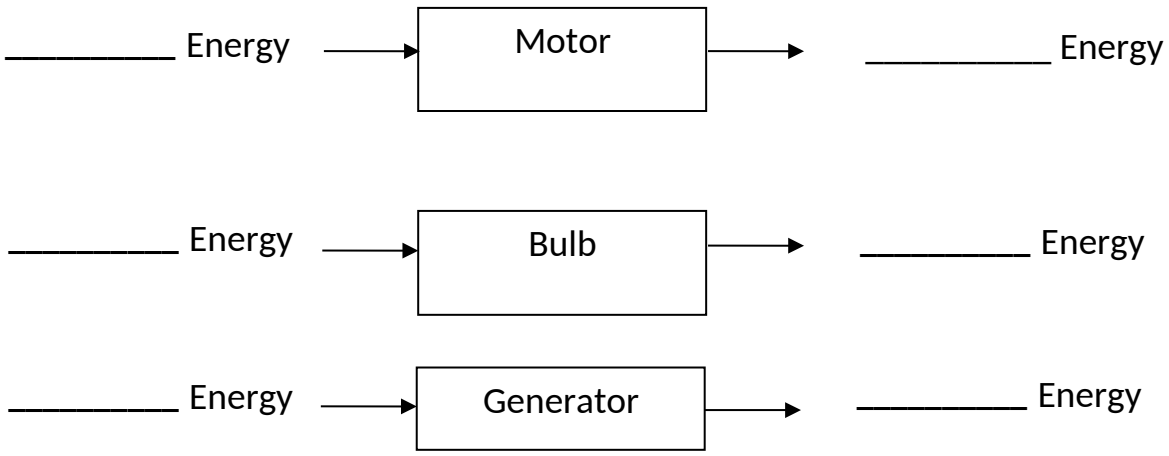
etc. In Technological Studies terms 'conservation of energy' has a different meaning. Instead it is looked upon as a rule. The Rule states that energy cannot be created or destroyed, but can only be changed from one form to another. It is 'transformed' or 'converted'. This rule is also termed a natural law.

The law of conservation of energy

The law of 'conservation of energy' states that for a closed system, where no energy goes in or out, the total energy within the system must always stay the same, although the form of energy may change. On the other hand, in an open system such as a power station, this rule leads to the conclusion that the total energy input to the system must be exactly equal to the total energy output. The extent to which the output energy is able to do useful work (of the desired type) is called the efficiency of the system. We can calculate this by comparing the useful output form the system with its energy input.

Energy Transformations

How energy can be transformed or converted is very important to a technologist. Some forms of energy is directly interchangeable (e.g. potential to kinetic), but others need to go through several changed to arrive to the desired form. Copy and complete these diagrams of simple transformations.



More energy transformations

Some systems will require more than one energy conversion to get the desired energy output. An example of this is converting geothermal energy into electrical energy.

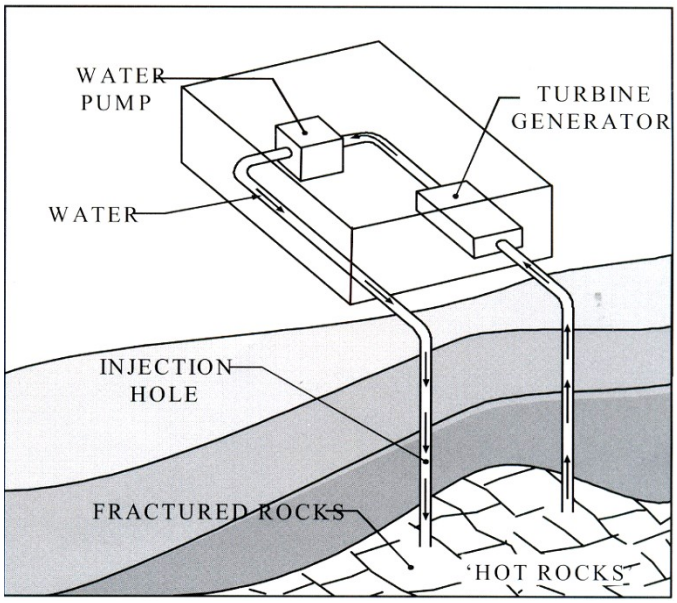
Geothermal Energy

The earth is a massive reservoir of natural heat energy. At its birth the earth was a molten mass at a temperate of 6000°C approx. The earth has been cooling down ever since and will continue to do so for

millions of years. In some places, the natural heat energy reaches close to the surface. Occasionally the molten materials cause the heat escape and this creates volcanoes. It is also known to heat up underwater reservoirs to create hot springs and geysers.

If this geothermal energy could be harnessed at temperatures of 250°C or higher, it could be used to create electricity, the heat being used to turn water into high-pressure steam to drive turbine generators. To achieve these temperatures however it would be necessary to drill deep into the earth's surface to reach "hot-rocks". In Britain it would be necessary to drill as deep as 6000 metres (4 miles approximately) to reach useful temperatures.

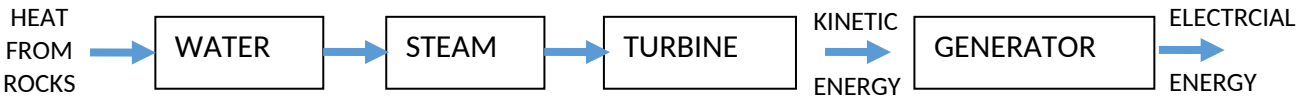
This is one method of collecting geothermal energy for transformation into electricity. This system consists of 2 boreholes, which penetrate the Earths crust and bore into a region of hot rocks. During construction an explosive charge would be detonated at the bottom of the injection hole to fracture the rocks.



When operating, water would be forced down this hole under pressure. It would then penetrate the hot rocks, pick up the heat, and return to the surface through a second borehole as steam. This stream would then be released through a pressurised system to drive the turbine generators.

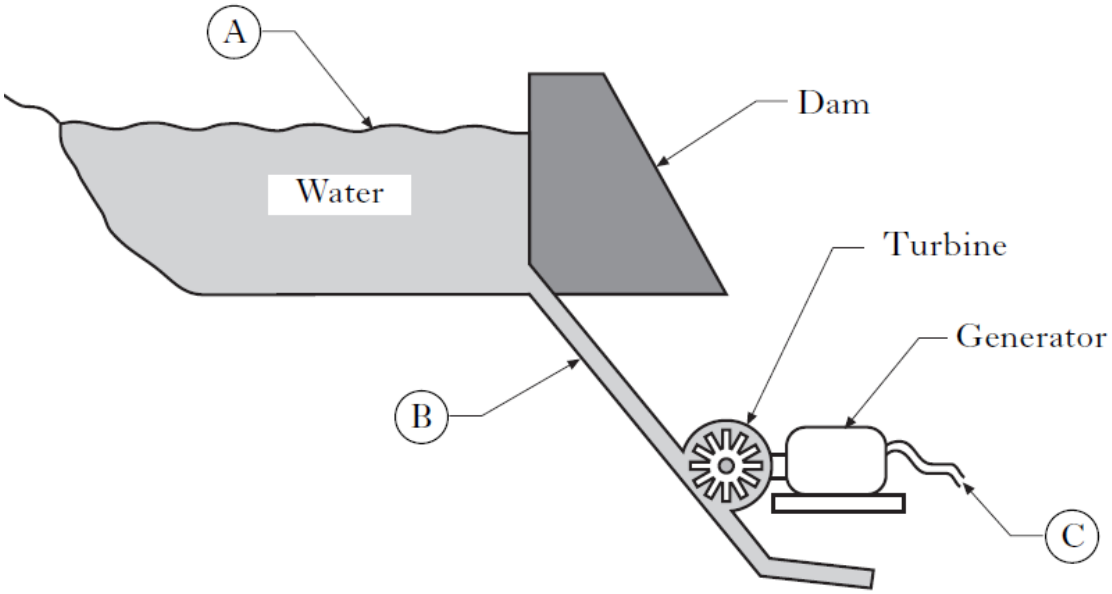
The energy conversion could be described as follows:

The heat energy in the rocks is transferred to the water in the injection hole pipe. This hot water is changed to high-pressure steam and transported to the surface. The heat energy from the steam is used to turn the blades of a turbine, producing rotational kinetic energy, which is used to create electricity from a generator.



A simplified hydroelectric power station is shown

State the **main** form of energy changes at positions A B and C.



A:

B:

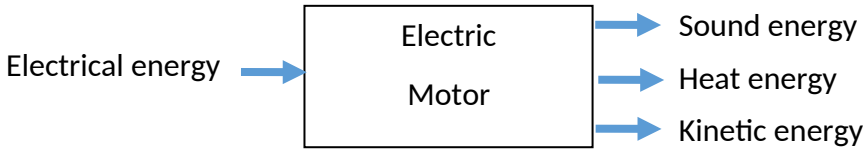
C:

Energy 'Losses' during transformation

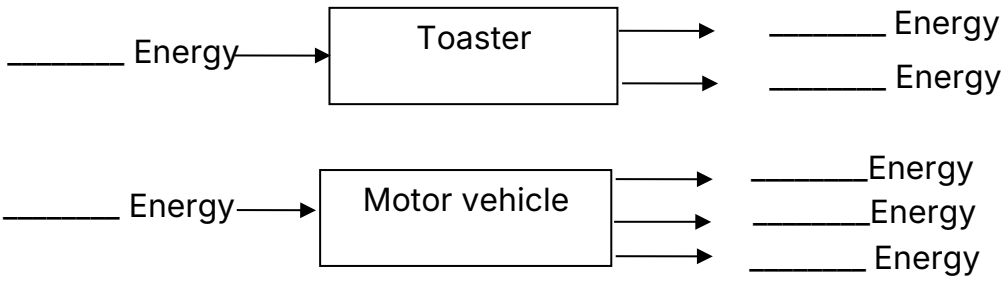
Although we have stated that energy cannot be destroyed, and that the energy output from a system is equal to its input, not all the energy is used efficiently. When an energy conversion takes place there is always an energy change that we do not desire – usually in the form of heat sound or friction from the moving parts of the mechanism. If we look at the simple energy conversions from earlier we can expand these to show the waste energy, or energy losses.

Example

An electric motor converts electrical energy to kinetic energy, but while doing so also creates sound energy and heat energy.



36. Complete these diagrams showing all the energy losses.



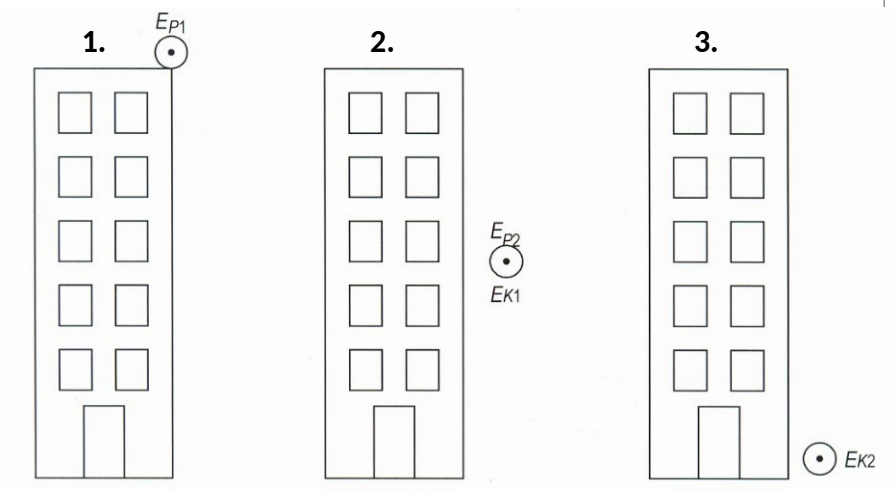
Calculating energy transfers

It has already been stated that energy cannot be created or destroyed – it can only be transformed to another energy. During an energy transformation, the total energy contained within any closed system remains constant. Knowing the total amount of energy at the start (or end) of any energy transformation tells us the total energy at any given time during the transformation.

Example

1. Consider a ball that is released from a high building. When the ball is at the top of the building, it has gravitational potential energy (E_{P1})

The total energy, $E_T = E_{P1}$



2. When the ball is released, it falls and some of the potential energy (E_{P2}) is converted into kinetic energy (E_{K1}). The total amount of energy (E_T) the ball possesses at this time is equal to the potential energy plus the kinetic energy.

The total energy $E_T = (E_{P2} + E_{K1})$

3. Just as the ball hits the ground, it no longer has any potential energy – all the potential energy (E_{P1}) has been converted into kinetic energy (E_{K1}). The total amount of energy now consists of the kinetic energy alone.

Total energy $E_T = E_{P1} = (E_{P2} + E_{K1}) = E_{k2}$

Worked example: transfer of energy (potential to kinetic)

A body of mass 30kg falls freely from a height of 20 metres. Find its final velocity and kinetic energy at impact. First calculate the initial potential energy.

$E_P = mgh$

$= 30 \times 9.8 \times 20$

$= 5880J$ $= \underline{5.9\ kJ}$

This potential energy is converted or transferred into kinetic energy, which means that the kinetic energy at impact is equal to 5.9 kJ. But this is not the end of the calculation!

To calculate the final velocity of the body we begin by taking E_k = 5.88 kJ.

$E_K = \frac{1}{2}mv^2$

$5.88 \times 10^3 = \frac{1}{2} \times 30 \times v^2$

$5880 = 15 \times v^2$

$v^2 = 5880 / 15$

$v^2 = 392$

$v = \underline{19.79\ m/s}$

This type of calculation can be completed for any type of energy conversion. Knowing the total amount of energy at any given time (beginning/middle/end) is the same at any other given time.



37. At the top of a 50cm ramp a 1.5kg buggy rolls down to the bottom using gravity. Work out the potential and kinetic energy at the top and middle of the hill. Calculate the transfer of energy.



38. At the top of a 80cm ramp, a 3kg buggy rolls down to the bottom using gravity. Work out the potential and kinetic energy at the top and middle of the hill.



39. A 5kg mass is raised steadily through a height of 2m. What work is done and what is the body's potential energy relative to the start?



40. A mass of 20kg is allowed to fall freely from a certain height above a datum. When the body is 16m above the datum, it possesses a total energy of 3531J. What is the starting height of the object?

Energy efficiency

It is possible to look at how well an energy system is operating by calculating its efficiency. The efficiency of an energy transformation is a measure of how much the input energy appears as useful as the output energy. This can be calculated using the following equation:

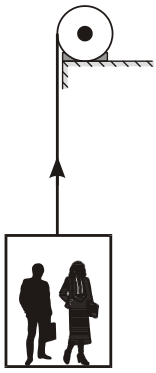
$$\eta = \frac{E_{out}}{E_{in}}$$

$$\text{Efficiency} = \frac{\text{Useful energy output}}{\text{Total energy input}}$$

Note – **η** is the ratio of output to input energy. This can never be greater than 1. In order to convert efficiency (**η**) to a percentage, it has to be multiplied by 100.

Example

An electric lift rated at 110V, 30A raises a 700kg load to a height of 20m in 2 minutes. By considering the electrical energy input and the potential energy gained by the mass, determine the percentage efficiency of this energy transformation.



Remember – translate the time into seconds! (2 mins x 60s = 120s)

The energy going into the system is electrical:

$E_e = IVt$

$= 30 \times 110 \times 120$

$= \underline{396\ kJ}$

The output Energy is potential so:

$E_P = mgh$

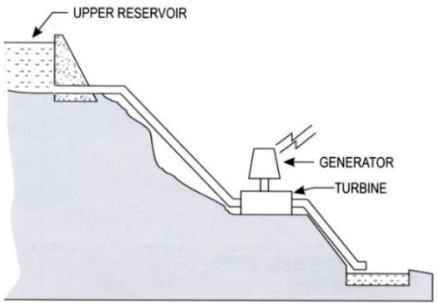
$= 700 \times 9.8 \times 20$

$= \underline{137.2\ kJ}$

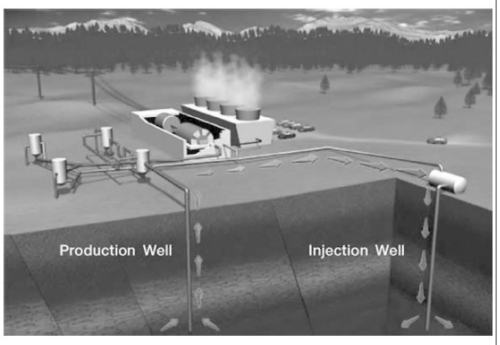
$$\eta = \frac{E\ out}{E\ in} = \frac{potential\ energy}{electrical\ energy} = \frac{137.2}{396} = 0.346$$

Percentage efficiency = 0.346 x 100 = <u>34.6%</u>

41. In a hydroelectric energy generating station, water is allowed to flow downhill through a turbine, which is connected to a generator. The water falls through a vertical height of 500m at a rate of 5000kg/s. If the energy transfer is 65% efficient, determine the amount of potential energy produced per second.



42. A Geothermal power plant uses the heat in the earth to help produce electricity.

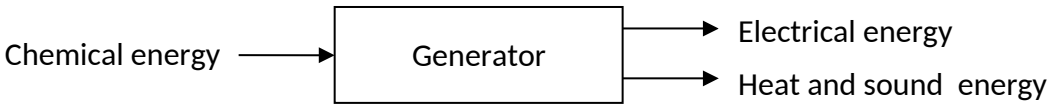


(a) Calculate the heat energy absorbed by 100 litres of water which is pumped into the earth at 10 °C and comes out as steam at 240 °C. (1 litre of water has a mass of 1 kg.)

(*b*) For every 15 MJ of heat energy that comes from the ground, the power plant produces 5.34 MJ of electricity. Calculate the efficiency of the power plant.

Energy audits

During energy transformations all the energy that goes *in* the system must come *out*. It is not possible to 'destroy' energy – it must go somewhere. Not all energy going into a system comes out as useful energy – this is called **waste energy**, or ***lost energy***. An example of this is that a light bulb turns electric energy into light energy, but it also creates heat – heat is the lost energy. Since we know that the total energy in any closed system must be constant, we can carry out meaningful calculations if we remember to take all types of input and output energy into account.



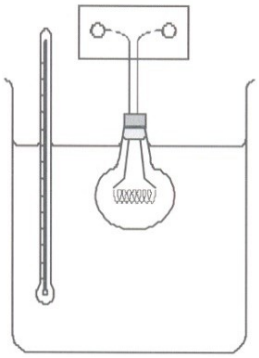
In the generator example above:

- the input energy is in the form of kinetic energy.
- the total output energy will be electrical energy, plus heat energy.

Hence, through conservation of energy $E_k = E_e + E_h$

In order to ensure that we have taken all energies into account, it is useful to carry out an **energy audit**. An energy audit is a list of all energies coming in and going out of a system. The total for the energies in **must** be the same as the totals for the energies out.

Worked example



In order to estimate the efficiency of an old filament electric light bulb, the bulb is immersed into a beaker of water as shown. Assuming all the heat energy is transferred to the water, it is possible to work out the efficiency of the bulb as a light producer.

Data:

Energy type going into the system is electrical (E_e) Power supply = 12 V

Current Drawn by the bulb = 5 A Volume of water in beaker = 0.5L

Initial temperature of water = 18°C Temperature of water after 10 minutes = 30°C

Time lightbulb is immersed for 10 minutes

Calculations:

t = 10 minutes x 60 seconds = 600s

Energy in (E_e) = IVt

= 5 x 12 x 600

= 36,000 J = **36 kJ**

Total Energy out is in the form of heat (E_h) and light (E_L)

$\Delta T = 30 - 18 = 12°C$

E_h out = $cm\Delta T$

= 4180 x 0.5 x 12

= 25,080 J = 25.1 kJ

The specific heat capacity (c) for water is 4180$Jkg^{-1}K^{-1}$

(This is also in the data booklet)

Useful energy out must equal the amount of energy in. However minus potential energy losses.

Energy in = 36 kJ Total energy out = 25.1KJ

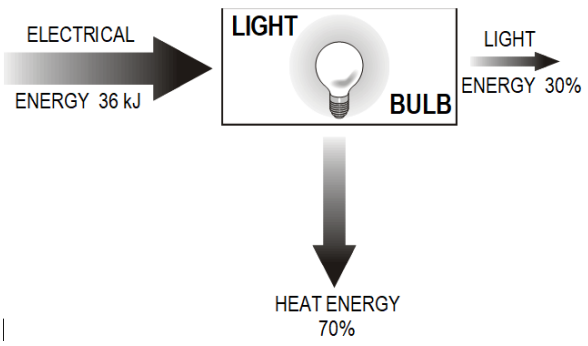
Useful energy output = Energy out – any losses ($E_{heat\ loss}$)

Therefore useful energy output = 36 – 25.1 = 10.9 kJ

Efficiency = $\frac{\text{Useful energy output}}{\text{Total energy in}} = \frac{10.9}{36}$ x 100% = 30% (of which 30% is only converted to light)

Waste energy (heat energy) 70% of input = 25.1kJ

This can be represented by a systems loss diagram opposite.

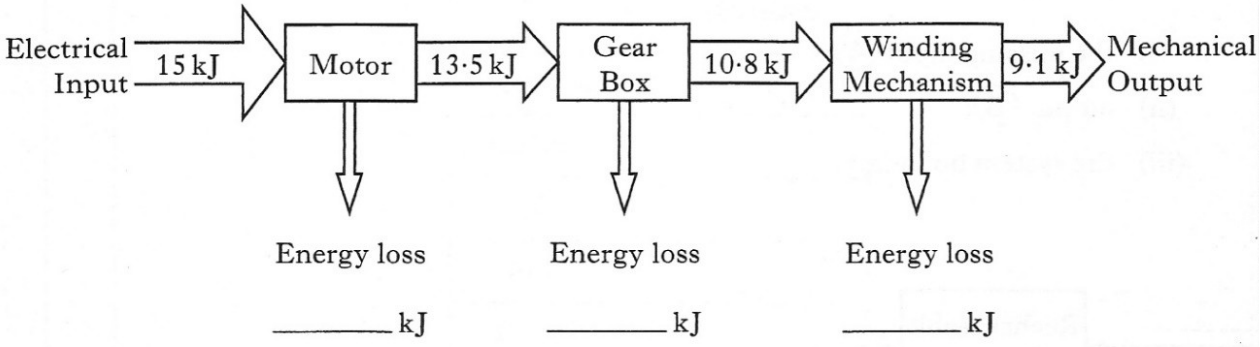


43. A deep fat fryer uses 2kJ of electrical energy per secc cooking oil from 20°C to 80°C in 6 minutes. If the s calculate the heat energy taken by the oil every secon

44. When full, a container of water holds 1.5 litres of water. The initial temperature of the cold water is 20 °C. A heater that operates from a 240V supply draws a current of 8A and has an efficiency of 75%. Determine how long it will take to heat the container of water to 90 °C.

45. An electric motor is used to drive a ski tow.

(*a*) Complete the energy audit below.



(*b*) Calculate the **total** energy **loss** in the ski tow.

Total energy loss = _____ kJ

(*c*) Complete the following sentence using the list of phrases given.

Input energy *Lost energy* *Output energy*

"Because the _____ is always greater than the _____ the efficiency will always be less than 100%."

d) *How efficient is the ski tow?*

46. A coffee machine heats one cup of water (0.14kg) from 18 °C to 89 °C in 16 seconds. The heating element operates from a 230V, 22A supply.

a) Calculate the electrical energy going into the coffee machine.

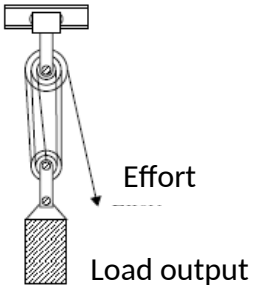
b) Calculate the heat energy transferred to the water.

c) Calculate the efficiency of the machine.

Extension: Energy of work done

1. When fully loaded, a fishing net has a mass of 500kg. Calculate the work done raising the fishing net 100m.

2. A weightlifter lifts a piece of gym equipment from the floor to arm's length, a vertical height of 1.2m. Calculate the mass of the equipment (in kg) if the minimum work done in lifting it is 600J.
3. In lifting an engine out of a car, a mechanic uses a block and tackle. How much work is done if a force of 500N is used in pulling the rope a distance of 4m?
4. A pulley lifting system designed to make it easier for an operator to lift a load onto a lorry is shown opposite. The operator applies an effort of 225N and pulls on 4.8m of rope to raise the load. Calculate the work done by the operator raising the load.



5. A mass of 50kg is raised to a height of 5m by a rope, which is wound around a pulley on a motor shaft of diameter 150mm. Determine the amount of work done by the motor.

Extension: electrical energy

1. A hot-air hand drier is activated for 30 seconds when the switch is pressed. The drier operates from a 230V supply and draws a current of 12A. Calculate the amount of electrical energy used when the drier is operating.
2. A 12 volt car battery has a capacity rating of 35 amp hours (that is, it can supply a current of 35 amps for 1 hour, or 5 amps for 7 hours, or any other combination giving an equivalence of 35 amp hours). Determine the amount of electrical energy the battery can supply.
3. A portable electrical generator can deliver energy at a rate of 6kJ per second. Calculate the current that can be drawn from the generator if the electricity is supplied at a voltage of 110 volts.
4. If the amount of electrical energy used by a 110V, 30A d.c. motor is 1.98MJ, for how long has the motor been in operation?
5. A turbine is used to drive a 110V electrical generator. If the output current is 3 amps, how much electrical energy does the generator produce in one hour?
6. An electric kettle takes 3.2 minutes to boil water. The original temperature of the water is 17°C and the kettle contains 1.8 litres of water. Calculate how much electrical current is used by the kettle from a 230 volt main supply.
7. The batteries on a prototype electric sports car is charged for 30 minutes from a main supply rated 230V 10A. Calculate the electrical energy used.

Extension: kinetic energy

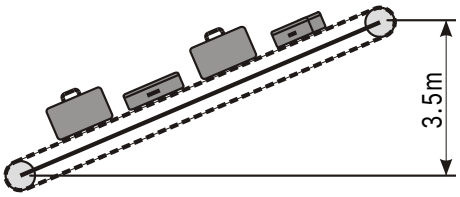
1. During a sheet-making process, 50kg ingots of metal are passed along rollers at a speed of 0.5 m/s.

Calculate the kinetic energy of each ingot.

2. A racing car drives around a circuit at a speed of 50 m/s. If the car has a kinetic energy of 500kJ, what is the mass of the car?
3. A girl of mass 50 kg is riding on her bicycle and has a kinetic energy of 2.5kJ. What speed is the girl moving at, and what is the kinetic energy of the bicycle if it has a mass of 30kg?
4. A crane raises a load of 200kg a height of 30 m in one minute at uniform speed. Determine the kinetic energy of the load when it is moving.
5. An escalator has six people on it with a total mass of 900kg. If the escalator moves at a uniform speed of 0.5 m/s, what is the average amount of kinetic energy that each person contains?
6. Since 1975 the US government has required that a container carrying radioactive material must be able to withstand a high speed crash with an impact velocity of 250 m/s. By comparison, the UK safety crash standards (in 1994) are carried out at only 50 m/s. If the container has a mass of 10kg then calculate the percentage increase in kinetic energy of the American test over the British test.

Extension: potential energy

1. Baggage handlers at an airport, place suitcases on to a conveyor belt, which lifts them up to the hold of the aeroplane as shown. Calculate the potential energy stored when a 20kg case is moved up the belt.



2. At what height must a drum of mass 100kg be suspended above the ground if it possesses 4 kJ of potential energy?
3. Calculate the potential energy available from a reservoir holding 1800 litres of water at a height of 260m. (1 litre of water has a mass of 1kg.)
4. Metal piles are driven into the ground using a pile driver. The driver is raised to a height of 5 m above the ground and then released. Calculate the weight of the driver if the potential energy stored when it has been lifted is 9810 joules.
5. If a steeplejack has a potential energy of 2000J when he scales a ladder to a height of 10m, what amount of potential energy will he possess at a height of 15m?

Extension: heat energy

C_{water} = 4180J/kg/°C C_{alum} = 899J/kg/°C C_{copper} = 387J/kg/°C $C_{mercury}$ = 138J/kg/°C

1. Calculate the heat energy required to heat 1.8kg of water from a temperature of 20°C until it begins to boil.
2. A hall has dimensions of 10 m × 328 m × 9 m. Assuming no heat losses, calculate the amount of heat energy required to raise the temperature of the air in the hall from 10°C to 20°C. The density of air is approximately 1.3 kg/m^3 and the specific heat capacity is 850 J/kgK.
3. The heating element in an electric shower can produce heat energy at a rate of 7 kJ/s. Water enters the system at a temperature of 15°C and is heated to 40°C. Estimate the flow rate (in litres per second) and state any assumptions you have made.
4. 54.3kJ of thermal energy is supplied to 1.3kg of oil having a specific heat capacity of 2.72kJ/kgK. If the initial temperature of the oil was 1.5°C, what will be its final temperature?
5. During a test on an engine, the following data were recorded below. If it is estimated that the cooling jacket of water absorbs 25% of the energy supplied by the fuel, calculate the quantity of water required per second in the cooling jacket.
    - Oil consumption = 3 kg/hour
    - Specific heat capacity of oil = 1360 J/kgK
    - Temperature rise in cooling water = 20°C
6. An electric heater supplies 13500J of heat energy to a metal block of mass 0.5kg. The temperature of the block rises from 20°C to 80°C during the heating process. Assuming that very little heat is lost from the block during the heating process, what is the specific heat capacity of the metal?
7. Calculate the energy required to raise the temperature of 750g of water from 34°C to 91°C.
8. When 6.3 x 10^5J of heat energy is added to a statue in a kiln oven, the statue's temperature rises from 16 °C to 95 °C. The specific heat capacity of the statue material is 3500J/Kg/C. Calculate the mass of the statue. Calculate the final temperature of 120g of water when 2.4 x 10^4J of heat is added to it. The initial temperature of the water is 19°C.
9. When 3kg of water is cooled from 80°C to 10°C how much heat energy is lost?
10. How much heat is needed to raise a 0.30kg piece of aluminium from 30°C to 150°C?
11. Calculate the temperature change when 10.0kg of water loses 232kJ of heat.
12. Calculate the temperature change when 1.96kJ of heat is added to 500g of copper.
13. 2.52 x 10^4J of heat is added to 2kg of mercury to reach a final temperature of 130°C. What was the initial temperature of the mercury?

Extension: power

1. A winch lifts a 500kg pallet of bricks to a height of 10m in a time of 15 seconds. Calculate the minimum output power of the lift.
2. An electric heater is rated at 3kW. Calculate how much heat energy the heater will deliver in one hour.
3. A car of mass 800kg is moving at a steady rate of 70km/hour. Calculate the kinetic energy of the car and the power output from the car in one minute.
4. An electric motor is rated at 10kW and runs from a main supply of 220V. If it consumes 10kJ of energy over a period of 30 seconds, what is the current used by the motor?

Extension: calculating energy transfers

1. A body of mass 30kg is projected vertically upwards with an initial velocity of 20 m/s. What is the initial kinetic energy of the body and to what height will it rise?
2. In a stamping machine, the die has a mass of 35kg and falls through a height of 2m on to a metal block. If the depth of indentation is 10 mm, find the average stamping force. Assume no rebound.
3. A boy stands on a skateboard at the top of a steep hill that has a height of 500m and allows gravity to increase his speed all the way to the bottom of the hill. If the boy has a mass of 60kg, calculate his potential and kinetic energy at the top, middle and bottom of the hill. (Ignore friction or any other resistance.)

Extension: efficiency

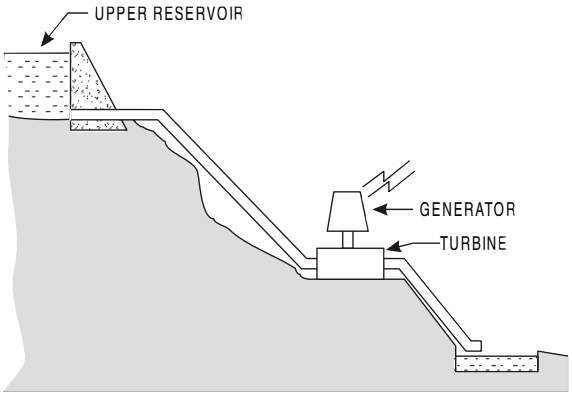
1. An electric kettle is rated at 240V, 10A. When switched on it takes three minutes to raise the temperature of 0.5kg of water from 20°C to 100°C. Determine:

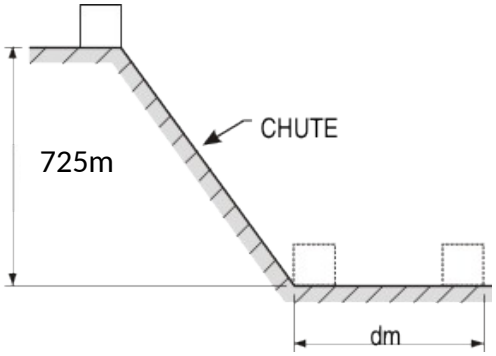
   (a) the electrical energy supplied in the three minutes

   (b) the heat energy required to raise the temperature of the water

   (c) the efficiency of the kettle.

2. In a hydroelectric electricity generating station, water is allowed to flow downhill through a turbine, which is connected to a generator. The water falls through a vertical height of 725m at a rate of 5,125kg/s. If the energy transfer is 72% efficient, determine the amount of electrical energy produced per second.





3. Boxes in a factory are transferred from one floor to another using a chute system as shown above. The boxes start from rest at the top of the chute and during the decent there is a 40% loss of energy. The boxes weigh 10kg each.

   a) Calculate the velocity of the boxes at the bottom of the chute.

   b) What is the distance *d* that each box will travel along the bottom floor before coming to rest if the frictional force opposing the motion of each package is 25% of its weight?

Extension: energy audits

1. A deep fat fryer uses 2kJ of electrical energy per second. It can raise the temperature of 1.5kg of cooking oil from 2°C to 80°C in 6 minutes. If the specific heat capacity of the oil is 2400 J/kgK, calculate the heat energy taken in by the oil every second and hence the efficiency of the fryer.
2. When full, a container of water holds 1.5 litres of water. The initial temperature of the cold water is 20°C. A heater that operates from a 240V supply draws a current of 8A and has an efficiency of 75%. Determine how long it will take to heat the container of water to 90°C.
3. A lift is used to raise a mass of 200kg to a height of 10m in two minutes. If the lift motor produces 12 MJ of energy in one hour, how efficient is the lift?
4. A vending machine, which dispenses hot drinks, heats up one cupful of water (0.15kg) from 30°C to 90°C. The heating element operates from a 240 volt supply and has a resistance of 12 ohms.

(a) Calculate the current drawn by the heating element.

(b) Calculate the heat energy transferred to the water.

(c) Calculate the time taken to heat the water if the system is 100% efficient.

(d) Calculate the percentage of energy lost if it actually takes 10 seconds to heat the water.

(e) Draw a system diagram showing the input to and outputs from the system. Include the percentages.

Extension questions

1. If an motor uses 500J to move a object for 30seconds what its power rating?
2. A 60W light bulb is left on for 20 seconds. How much energy has been used up?
3. If a 1kW electric fire has used 600J of energy how long has it been on for?
4. A car travelling at 12m/s has a force of 2kN applied to it by the engine. What is the power rating?
5. A microwave oven connected to the mains supply draws a current of 3A. What is its power rating?
6. A remote controlled car requires a current of 150mA to operate and is rated at 2W. How many 1.5V batteries would be required to make it go?
8. A microwave oven rated at 750W (Input) is 70% efficient. How long should it be switched on for if it has to supply 10kJ to your dinner?
9. An electric motor connected to the mains supply is rated as having an output power of 200W. If it is 60% efficient find the current drawn?
10. Three 100W bulbs are connected in parallel light a large room in a house. If the current drawn from the mains supply is 1.5A what is the average efficiency of the bulbs?

Revision homework 1: Understanding energy

1. Complete the following statement about energy:
Energy can be described as the ability to do …… .

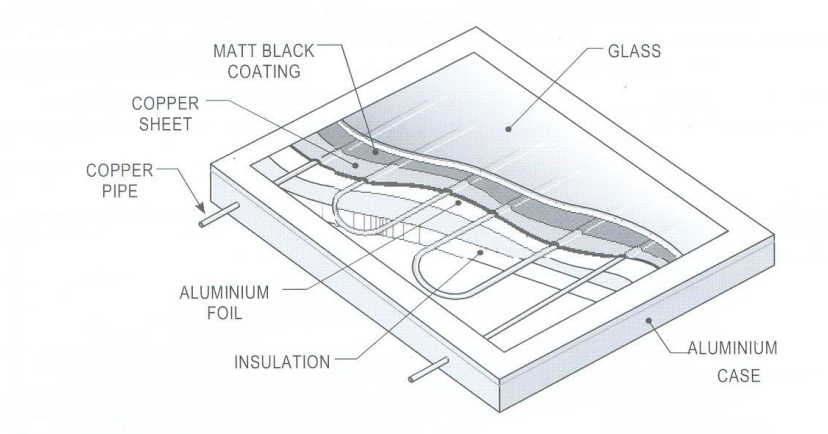
2. Where do all the earth's energy sources originally come from?

3. What is meant by the term 'fossil fuels', and what effect do they have upon 'Global Warming'?

4. Make a list of renewable energy sources that we can use to provide energy for the future.

5. What is the difference between a renewable (infinite) and a non-renewable (finite) source of energy?

6. Look at the diagram of a solar panel below and describe how it can be used to heat a building.



7 Why is it important to conserve energy?

8 Give an advantage and a disadvantage if using Nuclear Energy.

9 Describe some advantages and disadvantages of using wind and wave energy systems.

10 What are some of the ways that energy is wasted or lost in your home, or at school. Describe some possible solutions to this problem.

Revision homework 2

1. If 500J is used to move a object 30m, calculate the force that must have been applied.
2. A car is towed along a flat surface for 50m by a force of 30N then pulled up a slope for 20m with a force of 50N, calculate the total work done.
3. If 40J is used to lift a 1kg bag of sugar how high has it been raised? How high would it be raised on the moon where g= 1.6m/s2.
4. How much kinetic energy does a car of mass 500kg have if it is moving at 5m/s?
5. How much kinetic energy does a car of mass 500kg have if it is moving at 60km/h?
6. If a moving object has 20J of kinetic energy, find its velocity if it has a mass of 3kg.
7. If a 1kg mass is dropped from 2m, how fast is it travelling when it hits the ground?
8. If a 15kg object has a velocity of 4m/s when it hits the ground, how far has it fallen?
9. An electrical motor is connected to a winch system. If the motor is supplied by a 12V battery and draws a current of 0.5A for one minute calculate, how high it will raise a 10kg mass?
10. An electric motor is connected to a battery operated toy car. A 9V battery is used and supplies 150mA. If the car has a total mass of 500g calculate its speed after 15 seconds.
11. A builder carries 50kg of bricks up a ladder to a height of 3m. They use up 2kJ of energy, how efficient is are they?
12. A winch is powered by an electric motor that is rated at 5A. It is connected to the mains and is rated as being 80% efficient. How far will it move a weight that experiences 2kN of tractive resistance if the motor is turned on for 1 minute and turns at a constant speed?